

Course content and style

International Marketing Analytics



Dr Arne Floh, Senior Lecturer in Marketing
WU Vienna, Institute for International Business

arne.floh@wu.ac.at

[Office hours: book online \(Google Calendar\)](#)

WINTER TERM 2026/27



The journey begins





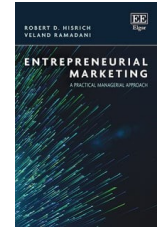
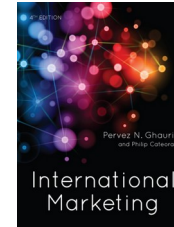
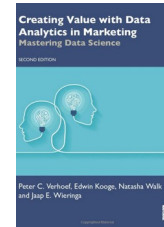
Personal and contact information

- **Senior Lecturer in International Marketing - WU Vienna**
- Deputy Program Director MSc ExInt
Export- und Internationalisierungsmanagement
- WWW: wu.ac.at/en/welthandel/arne-floh
- LinkedIn: <http://www.linkedin.com/arnefloh>
- T: 01 31336 6367
- E: arne.floh@wu.ac.at
- Room D1.5.068
- Office hours:
[book online \(Google Calendar\)](#)

Module Convenor - Dr Arne Floh



- Work Experience
 - International & Industry Background
 - Teaching Experience > 25 Years
- Teaching
 - International Marketing
 - International Business Projects
 - Marketing Research and Marketing Analytics
 - Global Branding
 - Supervision (BSc, MSc, PhD)
- Research
 - Dynamic Capabilities and Firm Performance
 - GenAI and Marketing
 - SMEs and Internationalization
- Private
 - Football
 - Electronic Gadgets
 - Austrian Wine





Marketing analytics is the measurement, management and analysis of marketing performance data to improve marketing effectiveness and return on investment. It turns data into decisions and shows where the next euro should go.



Key questions: how do price, place, product features and promotion drive performance measures such as sales, and how should a fixed budget be split across countries and channels?

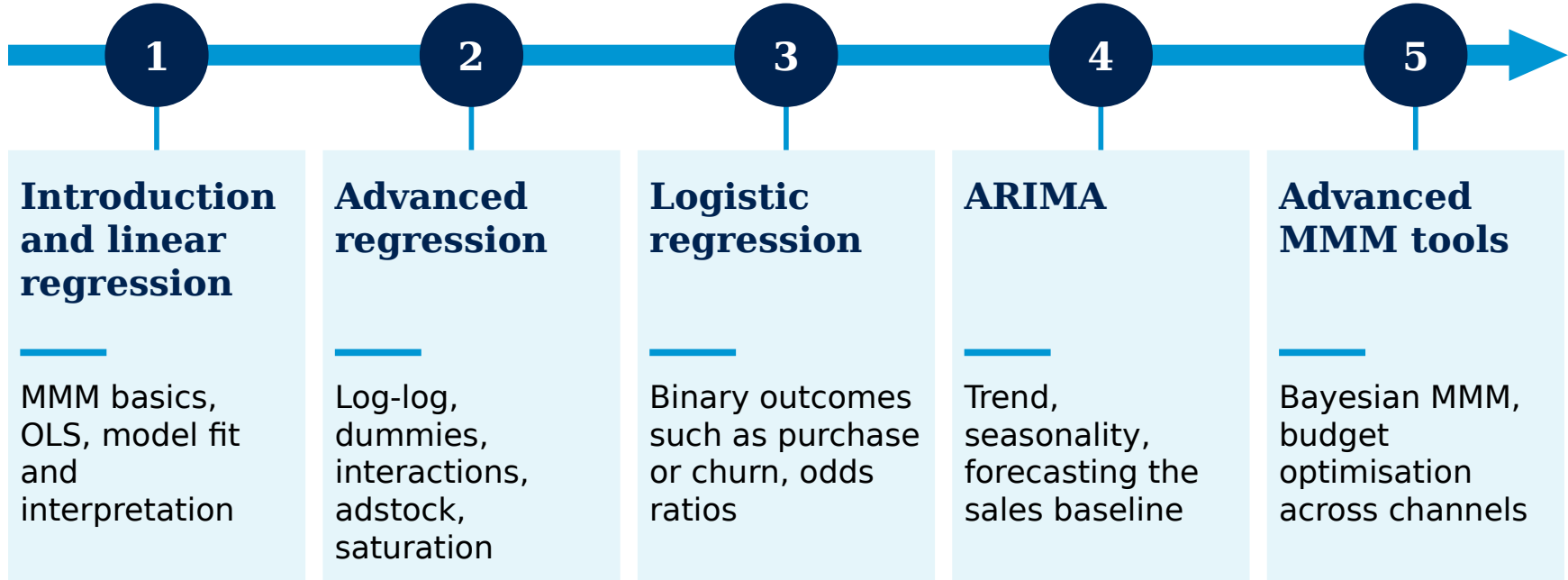


Tools: Python in Positron, GitHub for versioning and DataCamp courses; AI assistants do most of the typing, you check the results.



Case studies: Zalando and NEOH, plus a six-country teaching dataset with known answers for the labs.

Course roadmap: five sessions



Along the way: DataCamp certificates, coding exercises, guest speaker



At the end: written exam

Learning outcomes

On successful completion of this module, students will be able to:



Understand

- 1 **Explain the fundamentals of MMM**, its strengths and limits
- 2 **Describe real-world applications** of MMM in marketing



Apply

- 3 **Apply MMM in Python** to cross-sectional and longitudinal data
- 4 **Interpret results** and turn them into marketing decisions



Transfer

- 5 **Use AI tools** for analysis and reporting, and check the output
- 6 **Think critically** and work effectively in teams

Methods of teaching and learning

Every session

Concept lecture
60 min

Guided lab
90 min

Team exercise
60 min

Online quiz
not graded

Between sessions



- Read chapters and materials
- DataCamp courses and certificates
- Coding exercises
- Case study report in your group
- Q&A forum on Canvas



In the classroom



- Five in-person sessions
- Laptop with Python, Positron, GitHub
- AI-assisted coding, checked by you
- Guest speaker from industry



At the end: written exam, 60 minutes, paper and pencil, multiple-choice questions.

Module roadmap

Session	Topic	Methods
1	Introduction and linear regression	Course content and style; registration and set-up (Python, Positron, GitHub, DataCamp); MMM fundamentals; OLS, model fit and interpretation
2	Advanced regression	Log-log elasticities, dummy variables, non-linear effects, moderation and mediation, adstock and saturation, diagnostics
3	Logistic regression	Binary outcomes such as purchase or churn, odds ratios, model fit and classification
4	ARIMA	Trend, seasonality and autocorrelation; ARIMA and ARIMAX forecasts as the sales baseline
5	Advanced MMM tools	Bayesian MMM with PyMC-Marketing and Google Meridian; budget optimisation across channels and countries

Assessment strategy

20 %	20 %	30 %	30 %
 20 % Online certificates DataCamp, 4 × 5 % Individual	 20 % Coding exercises 4 × 5 % Individual	 30 % Written exam 60 minutes, multiple choice Individual	 30 % Case study report written report, groups of max. 4 Group

Grading scale (WU)

1 Excellent 90 % and above	2 Good 80 to 89 %	3 Satisfactory 70 to 79 %	4 Sufficient 60 to 69 %	5 Fail below 60 %
--------------------------------------	-----------------------------	-------------------------------------	-----------------------------------	-----------------------------

Attendance: at least 80 % of sessions. **AI tools:** expected; you own every number you report.

Software and set-up



Python 3.14

the language
for all analyses



Positron

the editor
where you
write, run and
see results



Quarto

turns code and
text into
reports and
slides



GitHub Desktop

gets the course
materials: clone
and pull



Posit Assistant

AI inside
Positron, with
GitHub Copilot

Set-up in Session 1: five steps

**Register
accounts**

**Install
the tools**

**Clone the
repo**

**Set up the
project**

**Run the
check**



Together in Session 1: bring your laptop and charger, no preparation needed.

Course repository: arnefloh-wu/international-marketing-analytics-teaching

4

DataCamp courses. Python courses assigned on Canvas, each ending with a certificate.

5 %

Per certificate. Upload each certificate on Canvas by its deadline.

20 %

In total. Together the four certificates count one fifth of the final grade.



Deadlines: the date for each certificate is on Canvas. The courses build the Python basics you need for the labs, so start early.

Coding exercises

4

exercises during the course

5 %

per exercise

20 %

of the final grade

How it works

**Guided lab
in class**

**Apply the code
to a similar
task**

**Submit on
Canvas**

**Feedback
and grade**

Rules



- Submit your Python or Quarto file; it must run from top to bottom
- AI assistants are allowed; add a short prompt log at the end
- Individual work: discuss ideas, but write and check your own code

Written exam

30 %

of the final grade

60

minutes exam time

1

exam at the end of
the course

MC

multiple-choice
questions

Format



- Paper and pencil, 60 minutes
- Multiple-choice questions
- Individual, closed book

Content



- All five sessions
- Concepts and interpretation
- Reading model output, not coding

Preparation



- Online quiz after every session (not graded)
- Session materials and book chapters
- Q&A forum on Canvas

Case study: written report

1

written report per
group

max. 4

students per group

30 %

of the final grade

30 Nov

2026, midnight:
deadline

What you submit on Canvas



report.qmd

your analysis and text
in one Quarto file that
renders without
errors



report.html

the rendered, self-
contained report



Appendix

AI prompt log and
contribution
statement, inside the
report

The task: answer four questions about Alpenglow's web shop with the methods of Sessions 1 to 4: what drives orders, price and advertising effects, subscriber churn, and the Q1 2026 baseline.

Tools: polars, plotnine, great_tables and statsmodels.

Case study: parts and points

A	What drives weekly orders? data check, first regression, model fit, assumptions	Session 1	15 pts
B	Price and advertising elasticities, temperature, adstock and saturation, moderation, mediation	Session 2	30 pts
C	Which subscribers will cancel? logistic regression, classification, retention-offer threshold	Session 3	20 pts
D	The Q1 2026 baseline for Germany ARIMA versus ARIMAX on a holdout, forecast with intervals	Session 4	20 pts
E	Recommendations and report summary, limitations, reproducibility, presentation	all sessions	15 pts

100 points, 20 criteria, each scored Excellent (90–100 %), Good (75–89 %), Adequate (60–74 %) or Insufficient (below 60 %).

Case study: submission and rules

**Pull course
repository**

**Copy into
your folder**

**Work in
the group**

**Render a
fresh copy**

**Upload to
Canvas**

Submission



- Mon 30 Nov 2026, midnight (Vienna)
- Canvas: one upload per group
- report.qmd + report.html, no zip
- Re-upload until the deadline

The report



- Max. 3,000 words, without code and tables
- Every number from code (inline code)
- great_tables and plotnine
- Appendix: prompt log, contributions



Two rules: if report.qmd does not render, reproducibility scores 0; a number in the text that does not match the code caps its criterion at Adequate.



VIENNA UNIVERSITY OF
ECONOMICS AND BUSINESS

**INSTITUTE FOR INTERNATIONAL
BUSINESS**

Welthandelsplatz 1, 1020 Vienna,
Austria

DR ARNE FLOH

Senior Lecturer in Marketing

F +43-31336-6367

arne.floh@wu.ac.at

www.wu.ac.at/en/welthandel/arne-floh

[Office hours: book online](#)